

Shadows Being Vacuumed Away: An MR Experience of Shadow Loss of Body with Spine-Chilling and Body-Trembling, and Shadow Loss of Thing

Ryu Nakagawa
Nagoya City University
Japan
nakagawaryu@sda.nagoya-cu.ac.jp

Kenta Hidaka
Nagoya City University
Japan
c215021@ed.nagoya-cu.ac.jp

Shimpei Biwata
Nagoya City University
Japan
c215022@ed.nagoya-cu.ac.jp

Sho Kato
Nagoya City University
Japan
c245709@ed.nagoya-cu.ac.jp

Taiki Shigeno
Nagoya City University
Japan
c215011@ed.nagoya-cu.ac.jp

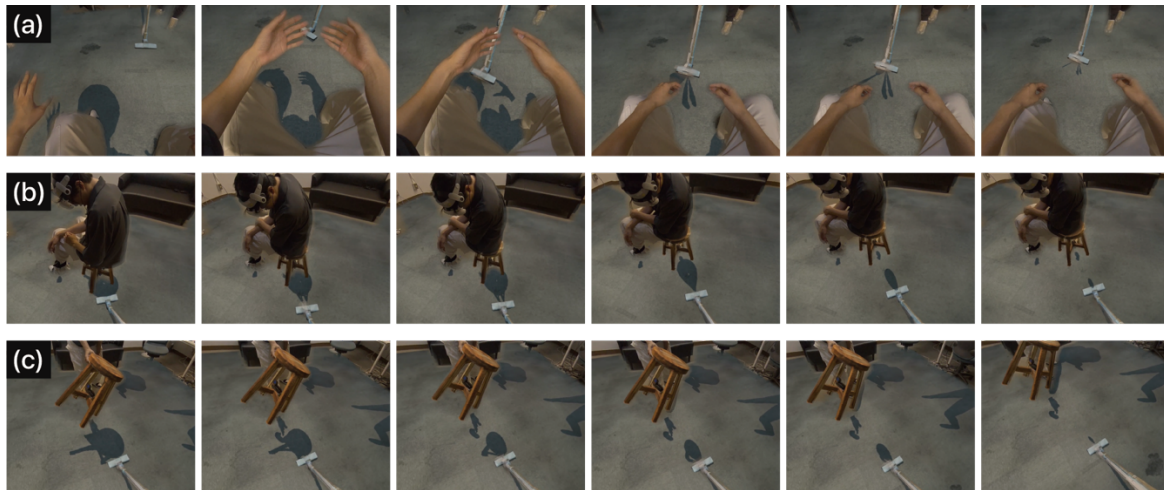


Figure 1: (a) A participant's shadow is being sucked away by a vacuum cleaner. (b) The shadow of the stool the participant is sitting on is being sucked away by a vacuum cleaner. (c) The shadow of the stool lifted by the participant is being sucked away by a vacuum cleaner.

CCS Concepts

• Applied computing; • Arts and humanities; • Media Arts;

Keywords

Virtual Shadow, Losing Shadow, Video see-through MR

ACM Reference Format:

Ryu Nakagawa, Kenta Hidaka, Shimpei Biwata, Sho Kato, and Taiki Shigeno. 2024. Shadows Being Vacuumed Away: An MR Experience of Shadow Loss of Body with Spine-Chilling and Body-Trembling, and Shadow Loss of Thing. In *SIGGRAPH Asia 2024 Posters (SA Posters '24)*, December 03–06,

2024, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3681756.3697937>

1 Introduction

Shadows play a crucial role in proving the existence of ourselves and objects in this world. In the research field of mixed reality, there have been numerous attempts to enhance the presence and realism of virtual objects by adding virtual shadows [Kolivand et al., 2014] [Naemura et al., 2002].

2 Motivation

Virtual shadows function as a means for virtual objects to gain a presence in the real world. Conversely, what if we were to erase the shadows of people and objects that already exist in reality? Many creators have likely considered this theme in the past. However, while it can be expressed in literature and film, making it a tangible, real-world experience has been challenging. In this work, we attempted to construct the strange experience of "losing shadows" by

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA Posters '24, December 03–06, 2024, Tokyo, Japan

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1138-1/24/12

<https://doi.org/10.1145/3681756.3697937>



Figure 2: (a) The scene during the actual experience. (b) The colored areas represent the virtual objects in Figure 2 (a). (c) The sections highlighted in orange, with rendering turned off (avatars and light source), generate virtual shadows.

leveraging the visual and auditory effects of the latest standalone Video see-through MR technology, combined with haptic stimuli to create a multimodal experience.

3 Overview of The Work

This artwork is a multi-participant MR experience using Meta Quest 3 headsets, Meta Quest Touch Pro Controllers (Pro Controllers), a customized haptic unit, and a customized vacuum cleaner and stool, where the shadows of a stool and the participants are sucked away by a vacuum cleaner operated by a facilitator (see Figure 1, 2). Additionally, when the participant witnesses their own shadow being sucked away, they hear synchronized sounds of the shadow disappearing, feel vibrations in their head and ankles, and experience a cooling sensation on the back of their neck. These haptic stimuli are generated using Linear Resonant Actuators (LRAs) and Peltier elements attached to the headset, as well as the vibration function of the Pro Controllers used to create the full-body shadow.

4 Implementation

4.1 Overview of The System

The system has been developed using Unity 2022.3.29f1, Meta XR Core SDK v65, Depth API v61, Final IK, Photon Fusion 2, among other tools. To represent the surreal situation of a shadow being sucked away, an MR space as shown in Figure 2 was constructed. Figure 2 (b) illustrates the areas within this MR space represented by virtual objects. Specifically, the floor, vacuum cleaner, and stool are overlaid with virtual objects created by 3D scanning them. A virtual light source is placed at the position of the actual light source, and avatars that follow the movements of the participant and facilitator generate virtual shadows in real-time on the virtual floor, which completely covers and occludes the real floor. The avatars are rendered in "Shadow Only" mode, with their visibility turned off so that only their shadows are cast. As a result, virtual shadows that move almost identically to the real shadows of the avatar's owner are drawn on the virtual floor. When a ray cast from a transparent object installed on the vacuum cleaner's head toward the virtual light source hits the transparent colliders embedded in the avatar, the vacuum cleaner's shadow-sucking action is triggered. Then, the avatar is sucked into the vacuum cleaner, making it appear as if the owner's shadow is being sucked away by the vacuum cleaner (see Figure 2 (c)).

4.2 Multi-Tracker System

Final IK is used to reflect the participant's movements onto the avatar and generate virtual shadows in real-time. By using the

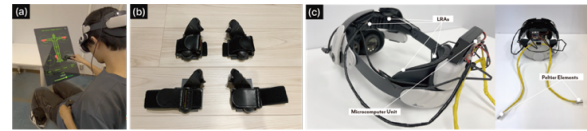


Figure 3: (a) Custom screen for assigning trackers and hands to make the participant's avatar follow. (b) Custom controller holders for Pro Controllers. (c) LRAs and Peltier elements attached to the head strap.

Simultaneous Hands and Controllers mode provided by the Meta XR Core SDK, it is possible to use controllers as motion trackers. However, typically, only two controllers can be used as trackers per player. Therefore, we devised a multi-tracker system that allows the controllers of multiple players in a multiplayer session to be consolidated and used by a single player. As shown in Figure 3 (a), by directly touching the hands or controllers to the setup panel, they can be registered as trackers for both hands and ankles. We created mounts using a 3D printer and generic belts to attach Pro Controllers as trackers to both hands and ankles (see Figure 3 (b)).

4.3 Generation of the Sensation of Shadow Suction from the Body

The strange sensation of a shadow being sucked away and lost was explored through a multimodal expression combining sound and haptic stimuli. The haptic stimuli included micro-vibrations on the head using LRAs, vibrations on the ankles via the Pro Controllers used for ankle tracking, and a cooling sensation on the back of the neck provided by Peltier elements. The haptic stimulus unit, which utilizes LRAs and Peltier elements, is attached to the BOBOVR M3 Pro head strap (see Figure 3 (c)).

5 Outlook

This artwork is still a work in progress, with further refinement needed for shadow suction and other aspects. The avatar can be adjusted to match the participant's height, but only one form is available. According to research by [Cortes et al., 2018] on the illusion of ownership between real bodies and virtual shadows, the acceptance of a virtual shadow decreases when its gender differs from the participant's own gender. Drawing on these examples, it would be valuable to investigate the impact of the gender and form of virtual shadows in MR environments.

References

- Hoshang Kolivand, Alhajhamad Hasan Zakaria, and Mohd Shahrizal Sunar. 2014. Shadow Generation in Mixed Reality: A Comprehensive Survey. In *IETE Technical Review*. 32, 3–15.
- Takeshi Naemura, Takuya Nitta, Atsushi Mimura & Hiroshi Harashima. 2002. Virtual Shadows in Mixed Reality Environment Using Flashlight-like Devices. *TVRSJ Vol.7 No.2*, 227–238.
- Guillaume Cortes, Ferran Argelaguet, Eric Marchand, and Anatole Lécuyer. 2018. Virtual shadows for real humans in a CAVE: influence on virtual embodiment and 3D interaction. In *Proceedings of the 15th ACM Symposium on Applied Perception (SAP '18)*. Association for Computing Machinery, New York, NY, USA, Article 1, 1–8.